

2025 ANNUAL GREENHOUSE GAS & CRITERIA AIR POLLUTANTS REPORT

Comprehensive Operational & Statutory Compliance Quantification

TOTAL SCOPE 1 & 2 EMISSIONS 1,908,885 tCO₂e -15.9% vs Baseline	TOTAL FLARED VOLUME 121.49 MMSm³ Lowest Flaring on Record	FLARING INTENSITY (HBNS) 0.93% of Gross Gas COMPLIANT (≤ 1.00% Decree 21-330)
MEASURED FLARING DRE 99.89% (HBNS) / 99.85% (ELM) VISR Camera Optical Verification	SALEABLE CARBON INTENSITY 33.72 kg CO₂e / BOE -6.2% YoY Decarbonization	NGSI METHANE INTENSITY 0.041 wt.% of Gas Top Decile Performance (< 0.20%)

Reporting Entity:	Groupement Berkine (Association Sonatrach / Occidental / Eni / TotalEnergies / Pertamina / Repsol)
Operating Assets:	Hassi Berkine South (HBNS, Block 404a) & El Merk (Block 208)
Reporting Period:	January 1, 2025 – December 31, 2025 (Annual Fiscal Cycle)
Verification Standard:	ISO 14064-1:2018 / GHG Protocol Corporate Standard / OGMP 2.0 (Level 4/5)
Statutory Regimes:	Algerian Executive Decree 21-330 (Gas Flaring) & Executive Decree 06-138 (Air Quality)
Calculation Basis:	API Compendium of GHG Emissions Methodologies (2021) / Site-Specific Gas Chromatography
Publication Date:	September 2026 Document Reference: GB-ENV-MRV-2025-01

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Technical Glossary of Industry & Regulatory Acronyms

API	American Petroleum Institute	MMSm³	Million Standard Cubic Meters
AR4 / AR5	IPCC Fourth / Fifth Assessment Report	NGSI	Natural Gas Sustainability Initiative
BOE	Barrels of Oil Equivalent (5,800 scf/bbl)	OGCI	Oil and Gas Climate Initiative
CAP	Criteria Air Pollutants (NO2, CO, SO2, PM, VOC)	OGMP 2.0	Oil & Gas Methane Partnership 2.0 (UNEP)
CPF	Central Processing Facility (HBNS & ELM)	SANGEA	API Greenhouse Gas Software Quantification Engine
DRE	Destruction & Removal Efficiency (Flame combustion)	VISR	Video Image Spectral Radiometry Camera
LDAR	Leak Detection and Repair (FLIR GFx320 OGI)	ZRF	Zero Routine Flaring by 2030 Initiative (World Bank)

EXECUTIVE SUMMARY: 2025 OPERATIONAL PERFORMANCE

1. 2025 Greenhouse Gas (GHG) Performance Highlights

- **Total GHG Reduction Achieved:** Consolidated Scope 1 + 2 emissions reached 1,908,885 tCO₂e in 2025, marking an overall reduction of **15.9%** relative to the 2.27 million-tonnes historical baseline (2021–2023 average).
- **Net Annual Decarbonization:** Net GHG emissions dropped by 3.5% in 2025 compared to 2024, expanding upon the 12.4% drop achieved in 2024.
- **Scope 1 Direct Reductions:** Gross Scope 1 emissions fell from 1.95 MMtCO₂e in 2021 to 1.59 MMtCO₂e in 2025, driven by extensive compressor sweep optimization and enhanced waste-heat utilization.

2. 2025 Methane (CH₄) Performance Highlights

- **Mass Methane Abatement:** Total methane emissions declined sharply from 7,124.8 tCH₄ in 2021 to **1,656.7 tCH₄** in 2025, representing a monumental **76.7% abatement** across Groupement Berkine assets.
- **Rigorous LDAR Campaign:** Successfully concluded the 2nd annual optical gas imaging (OGI) campaign utilizing high-flow sampler direct measurement for > 99% of detected fugitive leaks.
- **Measured Flaring Combustion Efficiency:** Incorporated continuous optical Video Image Spectral Radiometry (VISR) camera measurements confirming measured Destruction and Removal Efficiency (DRE) of **99.89% at HBNS** and **99.85% at El Merk**, vastly outperforming the generic 98.0% standard factor.

3. 2025 Flaring Performance & Regulatory Compliance

- **HBNS Flaring Slump:** CPF flaring volume at HBNS decreased by **24.9% YoY** in 2025, reaching 51.34 MMSm³ (lowest on record).
- **Statutory Compliance (Decree 21-330):** HBNS recorded an annual flaring intensity of **0.93% of gross gas produced**, achieving strict compliance under the mandatory **≤ 1.00% statutory ceiling** set by Article 9 of Algerian Executive Decree 21-330.
- **El Merk Performance:** El Merk recorded 70.15 MMSm³ flared (1.65% intensity). Front-End Engineering Design (FEED) for the low-pressure Flare Gas Recovery System (FGRS) is completed, with EPC commissioning targeted for Q3 2026 to ensure full statutory compliance.

Performance Category	2021 Baseline	2024 Actual	2025 Actual	5-Year Progress
Gross Scope 1 (MMtCO ₂ e)	1.947	1.662	1.592	-18.2% Reduction
Indirect Scope 2 (MMtCO ₂ e)	0.259	0.326	0.317	+22.4% Production Power
Total GHG (MMtCO ₂ e)	2.206	1.989	1.909	-13.5% vs 2021 (-15.9% vs Base)
Methane Emissions (tCH ₄)	7,124.8	2,703.0	1,656.7	-76.7% Abatement
Total Flared Volume (MMSm ³)	196.30	132.98	121.49	-38.1% Flaring Reduction
Carbon Intensity (kg/Total BOE)	17.46	14.81	14.58	-16.5% Efficiency Gain

STRATEGIC DECARBONIZATION ACTIONS & COMMITMENTS

1. Commitment to World Bank Zero Routine Flaring (ZRF) by 2030

Groupeмент Berkine is formally dedicated to eliminating routine gas flaring by 2030 across all production facilities. During 2025, major engineering milestones were realized:

- **Concept-Selection Approval:** Completed the concept-selection phase for advanced Flare Gas Recovery Units (FGRU) capable of recovering low-pressure flash gas from separation trains and routing it into the reinjection compression system.
- **FEED Execution:** Launched Front-End Engineering Design (FEED) for both HBNS and El Merk CPF flare collection networks, incorporating liquid knockout drums, variable-frequency screw compressors, and automated purge gas controllers.

2. Full Alignment with OGMP 2.0 Level 4/5 Quantification

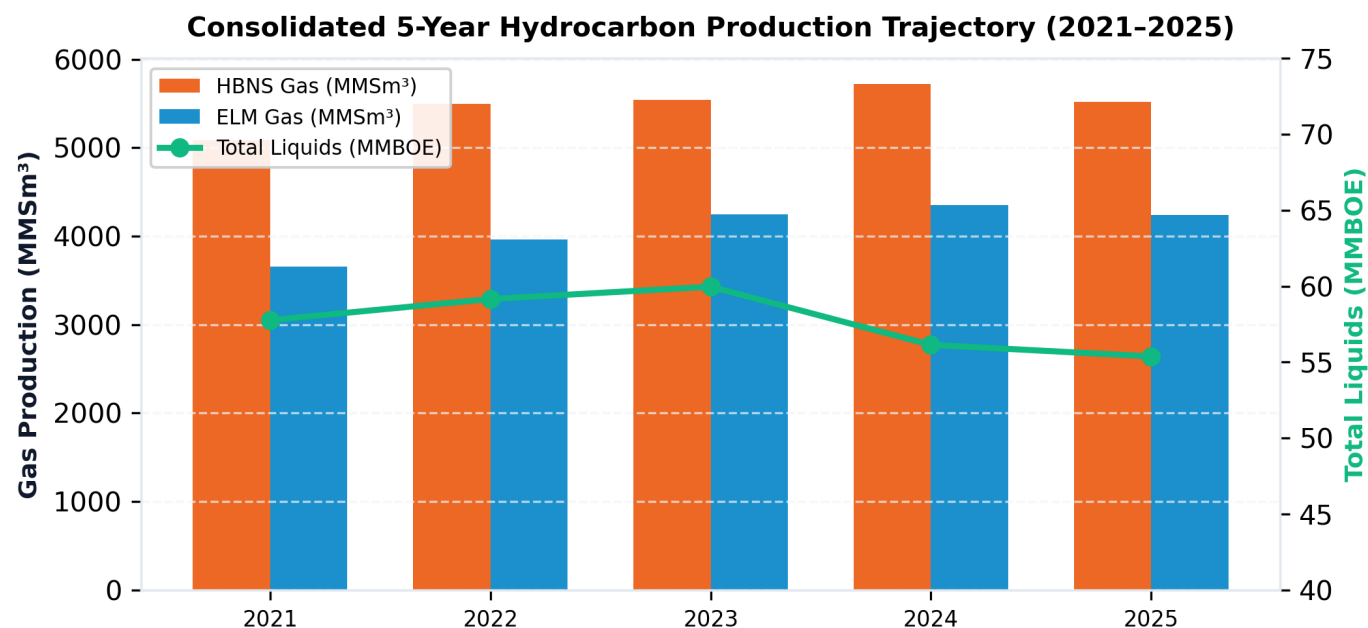
In support of Algeria's national climate pledges and partner corporate reporting guidelines, Berkine has adopted the **Oil & Gas Methane Partnership (OGMP 2.0)** Gold Standard framework:

- **Level 4 Site Measurements:** Direct measurement of source-level fugitive component emissions utilizing calibrated high-flow samplers and ultrasonic flowmeters, replacing generic API Compendium emission factor tiers.
- **Level 5 Top-Down Reconciliation:** Began technical evaluations for regional aerial LiDAR and satellite surveillance (Sentinel-5P, GHGSat) to perform top-down site-level verification and reconcile bottom-up component inventories.

3. Ongoing Capital Abatement Projects

Initiative / Project Title	Target Asset	Scope & Engineering Mechanism	Expected Impact
Flare Gas Recovery System (FGRS)	El Merk Complex	Compress LP flare gas into main booster re-injection train	-65,000 tCO2e/yr; Flaring < 1.0%
VISR Camera Continuous DRE	HBNS & ELM Flares	Continuous multispectral infrared combustion tracking	Accurate Level 4 flare CH4 MRV
High-Pressure Gas Re-injection	HBNS CPF	Upgrade cylinder valves on HP reciprocating compressors	-25,000 tCO2e/yr from venting
Quarterly LDAR Leak Repair	All Operating Units	Comprehensive FLIR GFx320 optical imaging + Hi-Flow	Zero fugitive leaks > 10,000 ppm
Dry Low NOx (DLN) Upgrade Study	El Merk Turbines	Burner retrofits on GE Frame 5/6 gas generation turbines	Decree 06-138 NO2 compliance

SECTION 1: YEARLY HYDROCARBON PRODUCTION PROFILE

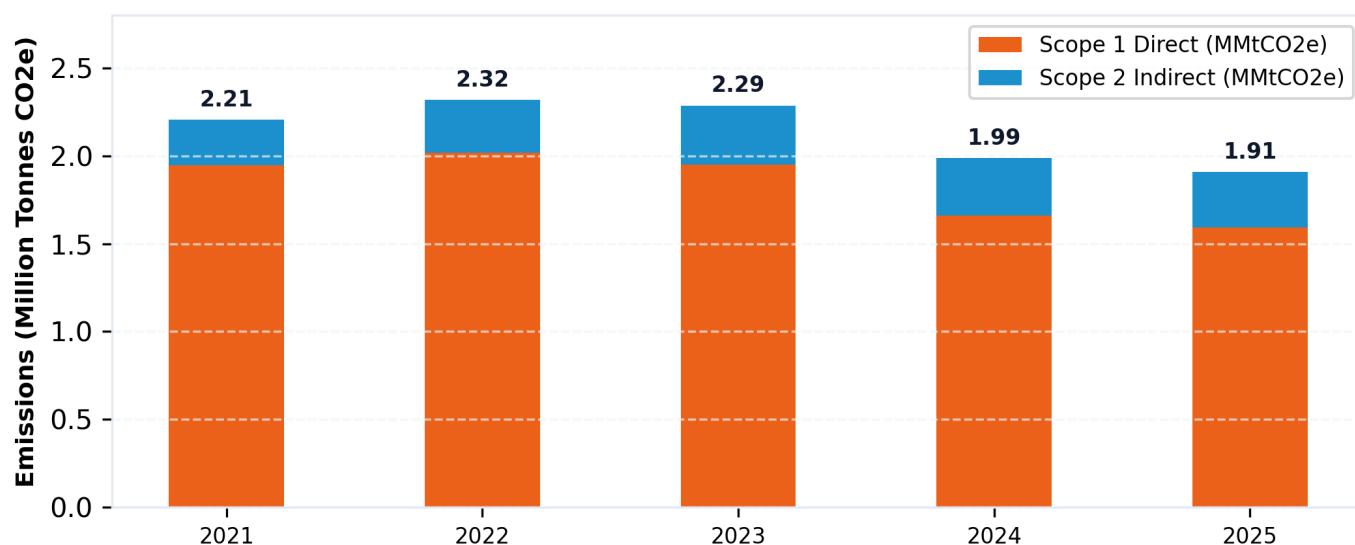


Hydrocarbon Production Dynamics (2021–2025):

Gross natural gas production across Groupement Berkine concessions remained resilient at **9,757.5 MMSm³** in 2025 (5,517.6 MMSm³ at HBNS; 4,239.9 MMSm³ at El Merk). Over **92.1% of all gross gas produced** (8,990.0 MMSm³) was reinjected into the TAGI and Strunian reservoirs to sustain reservoir pressure, driving total gross production to **130.93 MMBOE**. Saleable hydrocarbon production (crude oil, stabilized condensates, and LPG export) totaled **56.62 MMBOE**.

SECTION 2: TOTAL GHG EMISSIONS BY SCOPE

GB Consolidated Total GHG Emissions by Scope (2021–2025)

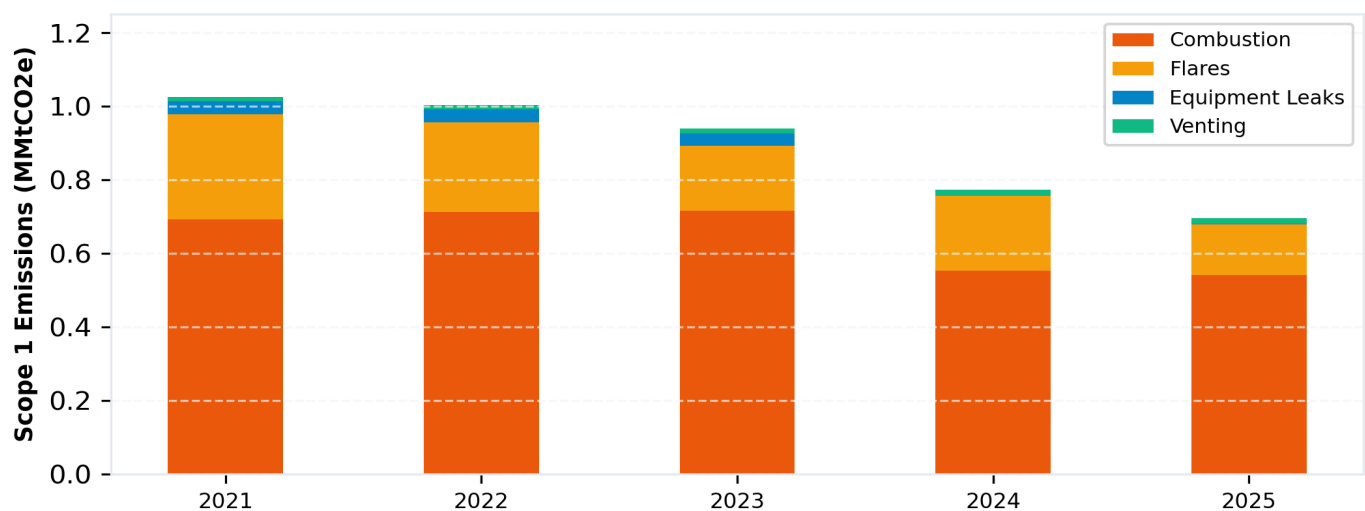


Quantified GHG Scope Breakdown (ISO 14064-1:2018):

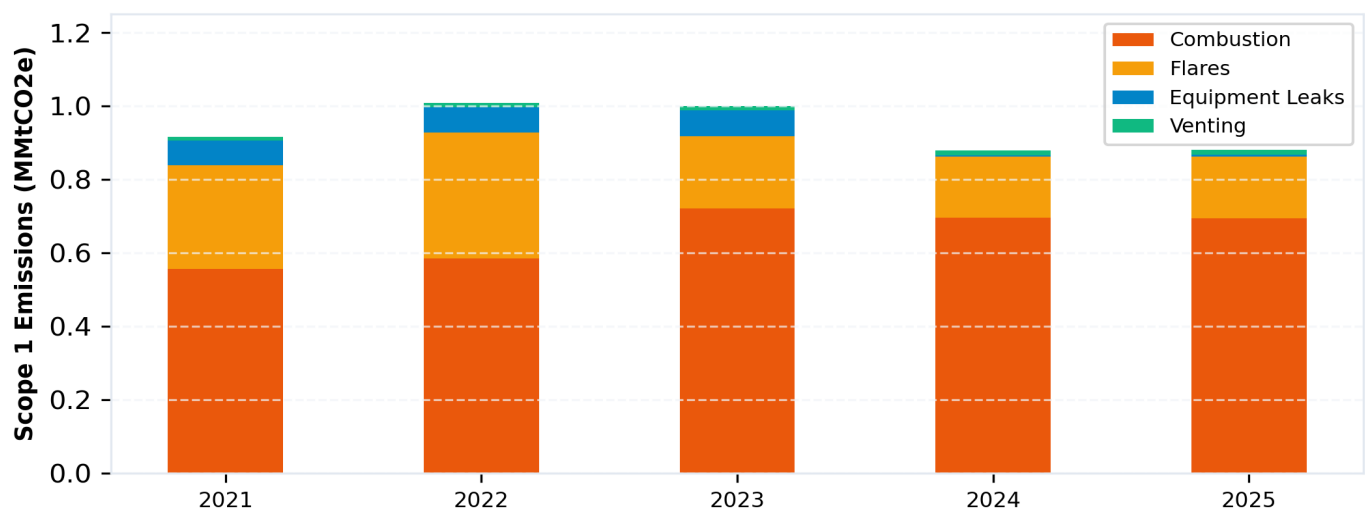
Total greenhouse gas emissions totaled **1,908,885 tCO₂e** in 2025, composed of **1,591,573 tCO₂e (83.4%) Scope 1 direct** and **317,313 tCO₂e (16.6%) Scope 2 indirect**. Over the 5-year accounting timeframe, direct Scope 1 emissions fell by **18.2%** (from 1.95 MMtCO₂e in 2021), proving the efficacy of continuous operational flaring cutbacks and combustion efficiency upgrades.

SECTION 3: SANGEA MODULAR GHG EMISSIONS (HBNS & EL MERK)

HBNS (Block 404a) SANGEA Modular GHG Emissions Breakdown



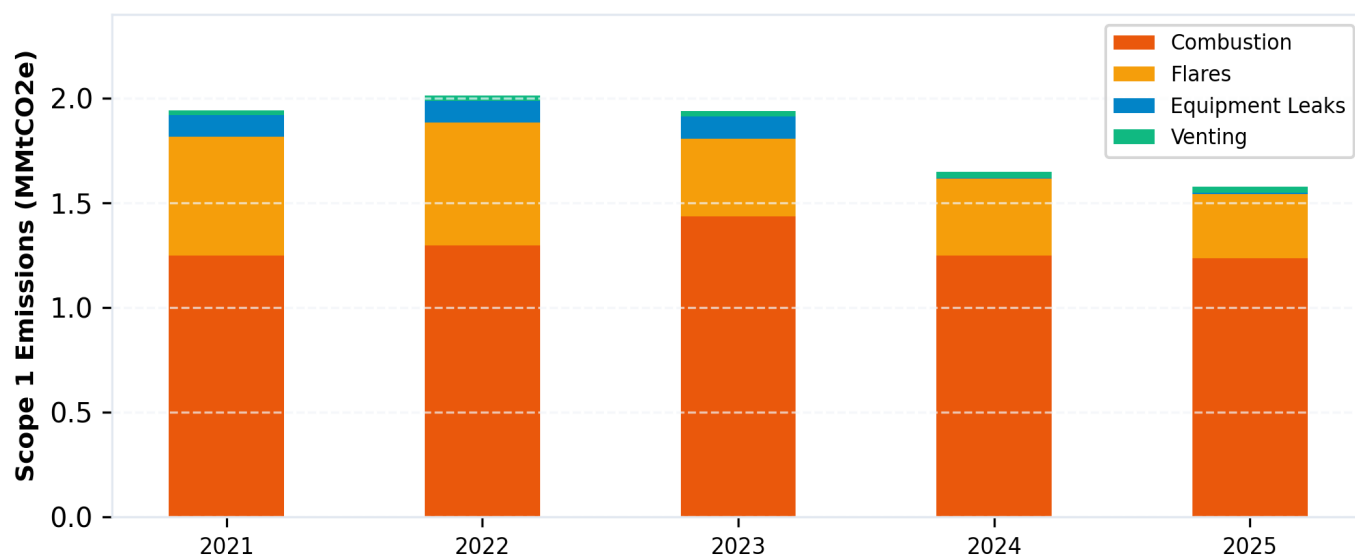
El Merk (Block 208) SANGEA Modular GHG Emissions Breakdown



Modular Emission Profile Disaggregation: Stationary combustion in high-pressure gas turbines and process heaters represents the primary emission source across both central processing facilities (76.6% of HBNS Scope 1; 78.4% of El Merk Scope 1). Flaring represents the second largest modular category, which has declined dramatically at HBNS from 284,807 tCO₂e in 2021 to 138,248 tCO₂e in 2025.

SECTION 3 (CONT.): CONSOLIDATED GB MODULAR EMISSIONS

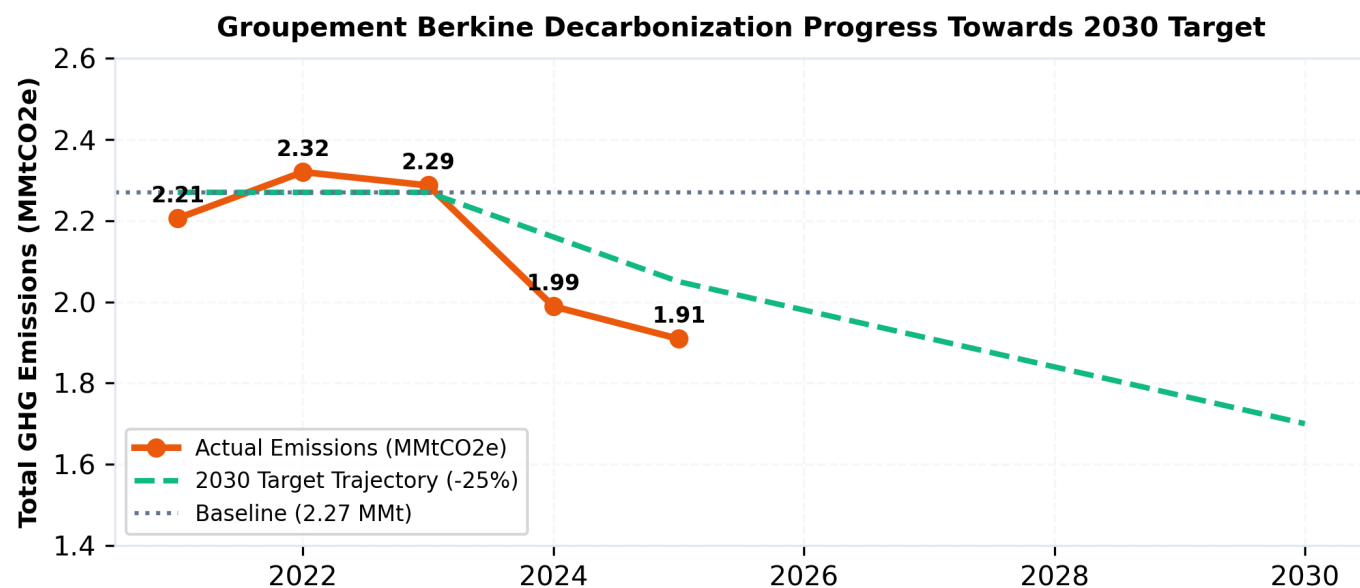
GB Consolidated SANGEA Modular GHG Emissions Breakdown



Groupement Berkine Total Modular Footprint:

In 2025, consolidated stationary combustion accounted for **1,234,795 tCO₂e**, operational flaring generated **306,015 tCO₂e**, while fugitive equipment leaks and venting contributed **7,972 tCO₂e** and **28,364 tCO₂e** respectively. Fugitive emissions have experienced an extraordinary **92.3% decrease** since 2021 due to the comprehensive quarterly LDAR repair campaigns.

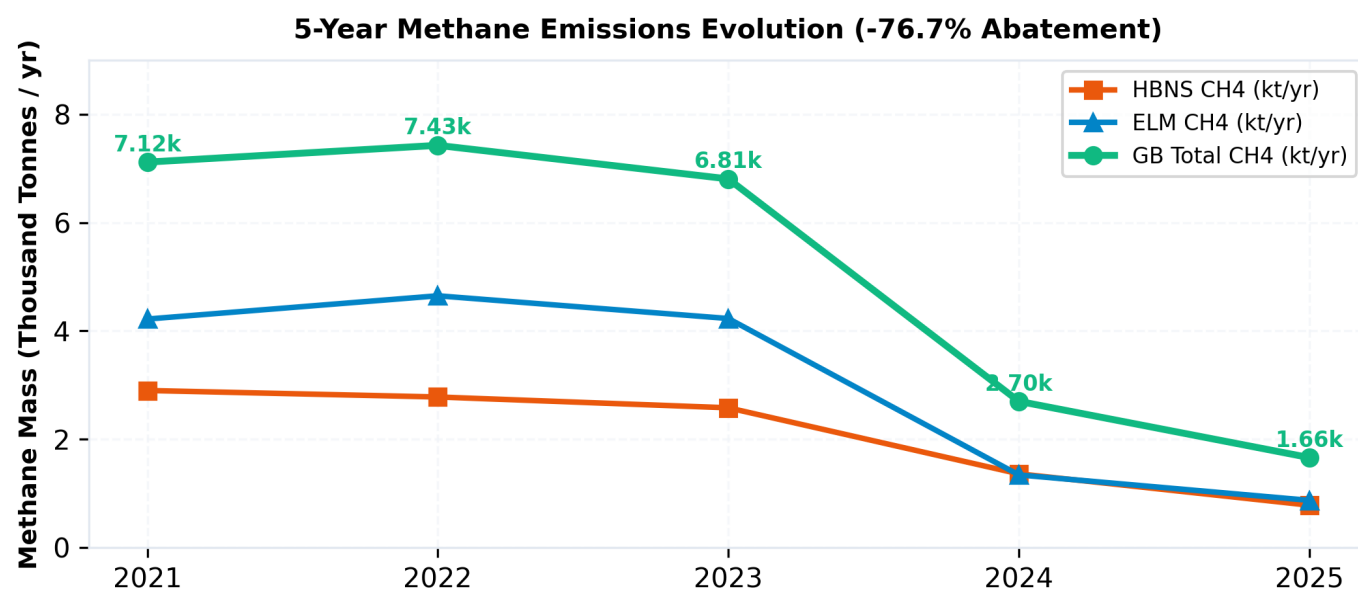
SECTION 4: 2030 TARGET PROGRESS & TRAJECTORY



Tracking Corporate Net Decarbonization Targets:

Groupement Berkine established a corporate milestone to reduce annual gross emissions from the 2.27 MMtCO₂e baseline to **1.70 MMtCO₂e by 2030 (-25%)**. With 2025 verified emissions reaching 1.91 MMtCO₂e (-15.9%), the consortium is tracking **comfortably ahead of the required linear trajectory**, with upcoming flare gas recovery projects expected to close the remaining gap.

SECTION 5: YEARLY TOTAL METHANE (CH₄) EMISSIONS

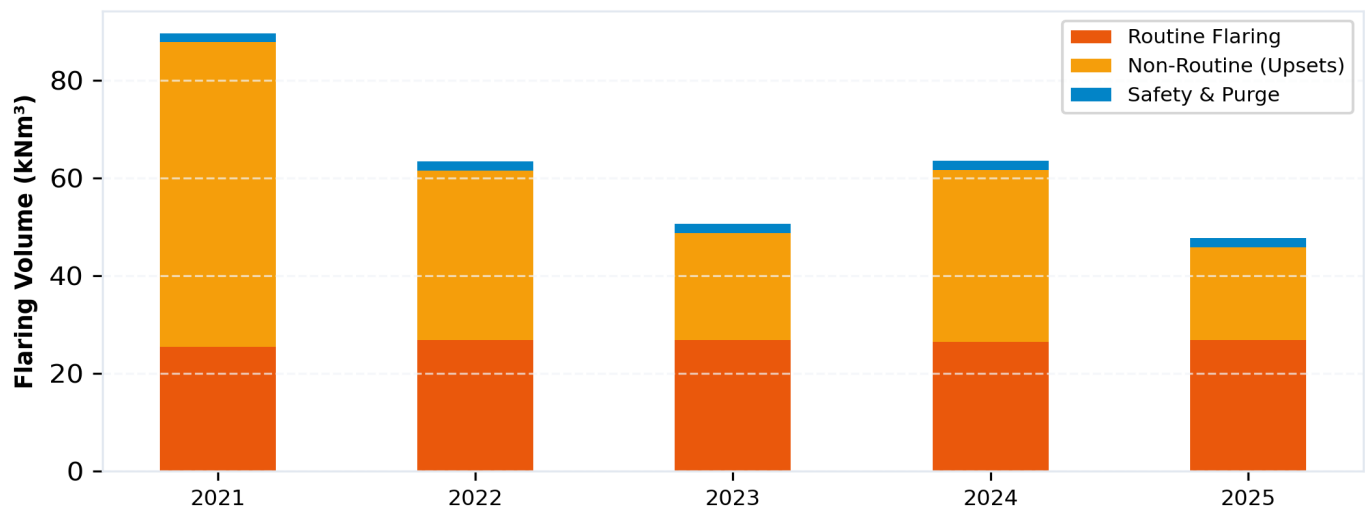


Methane Abatement Performance (-76.7% Since 2021):

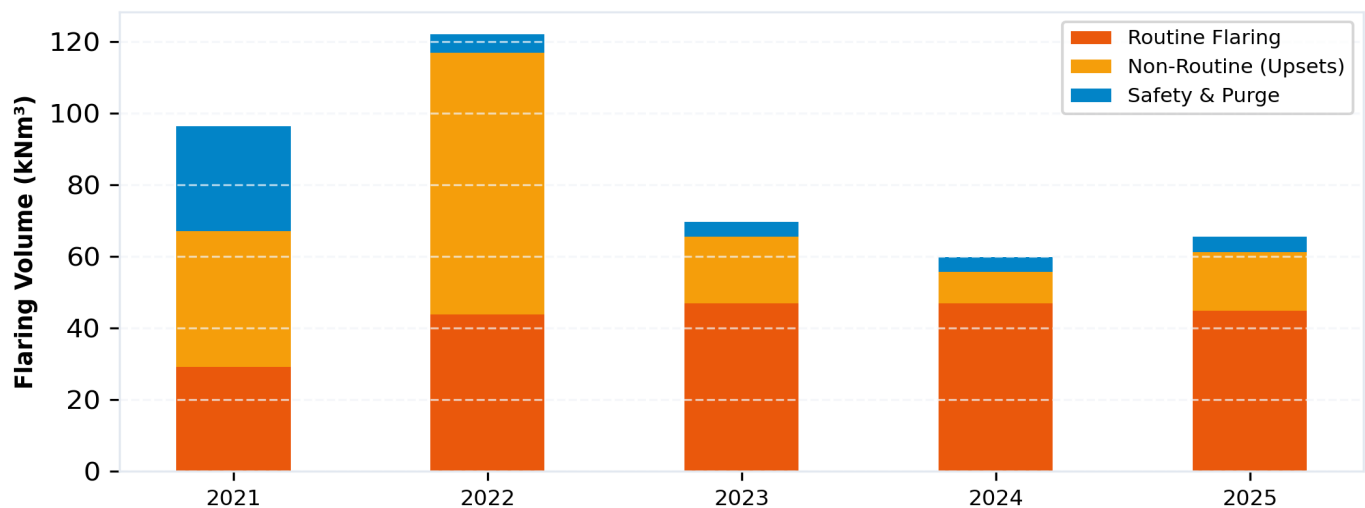
Consolidated methane emissions dropped from **7,124.8 tCH₄ (178,120 tCO₂e)** in 2021 to **1,656.7 tCH₄ (46,387 tCO₂e)** in 2025. HBNS emitted 782.6 tCH₄, while El Merk recorded 874.1 tCH₄. This rapid reduction was driven by optical camera LDAR leak eliminations and empirical VISR measurements verifying flare flame destruction efficiency exceeding 99.85%.

SECTION 6: OPERATIONAL CPF FLARING BREAKDOWN

HBNS CPF Flaring Streams Disaggregation (kNm³)



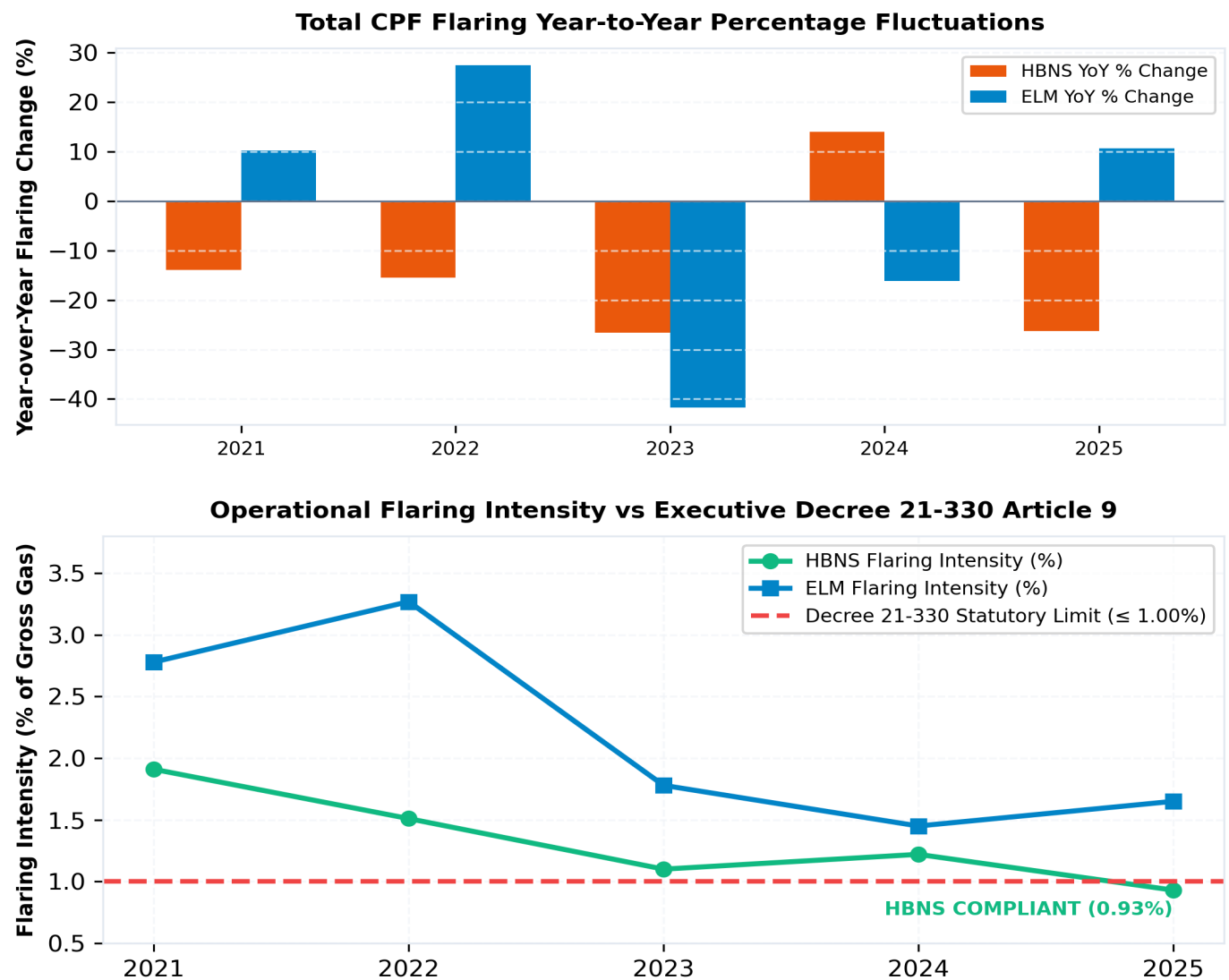
El Merk CPF Flaring Streams Disaggregation (kNm³)



CPF Flaring Stream Analysis (kNm³):

At HBNS, routine flaring stabilized at 26,820 kNm³ (56.2% of CPF flared gas), while non-routine flaring fell to 18,972 kNm³. At El Merk, routine flaring was 44,787 kNm³, non-routine flaring was 16,370 kNm³, and safety/purge gas was 4,226 kNm³. The impending commissioning of the FGRS will recover the routine streams and re-inject them directly into the reservoir.

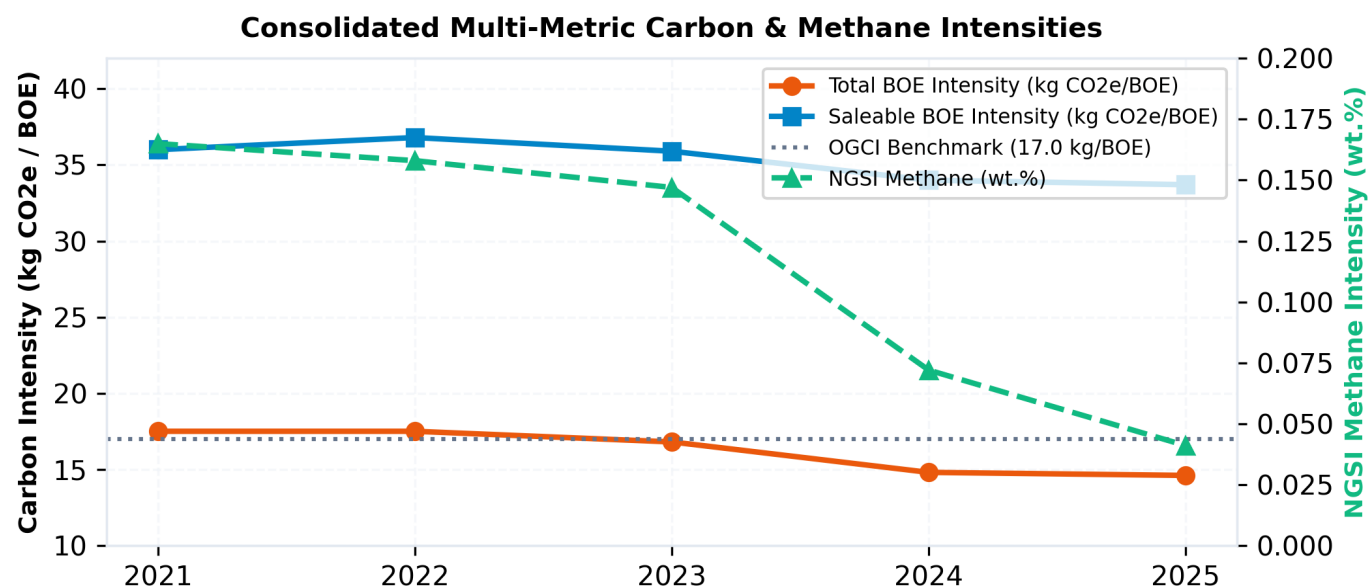
SECTION 7: FLARING COMPLIANCE (DECREE 21-330)



Statutory Verification Against Executive Decree 21-330 Article 9:

Article 9 of Decree 21-330 strictly mandates that operational flaring shall not exceed 1.00% of gross gas production. In 2025, **HBNS achieved full statutory compliance with a flaring intensity of 0.93% (51.34 MMSm³ flared / 5,517.6 MMSm³ gas produced)**. El Merk recorded 1.65%, maintaining active capital remediation plans to bring intensities under 1.00% by Q3 2026.

SECTION 8: MULTI-METRIC PERFORMANCE INTENSITIES

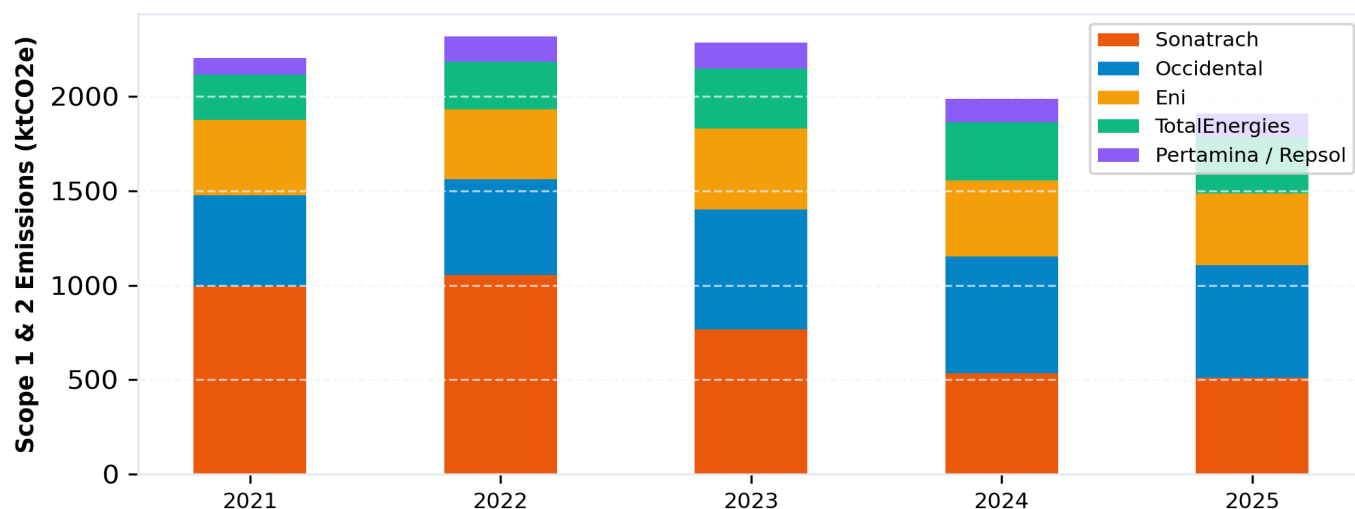


Dual-Denominator & Methane Intensities:

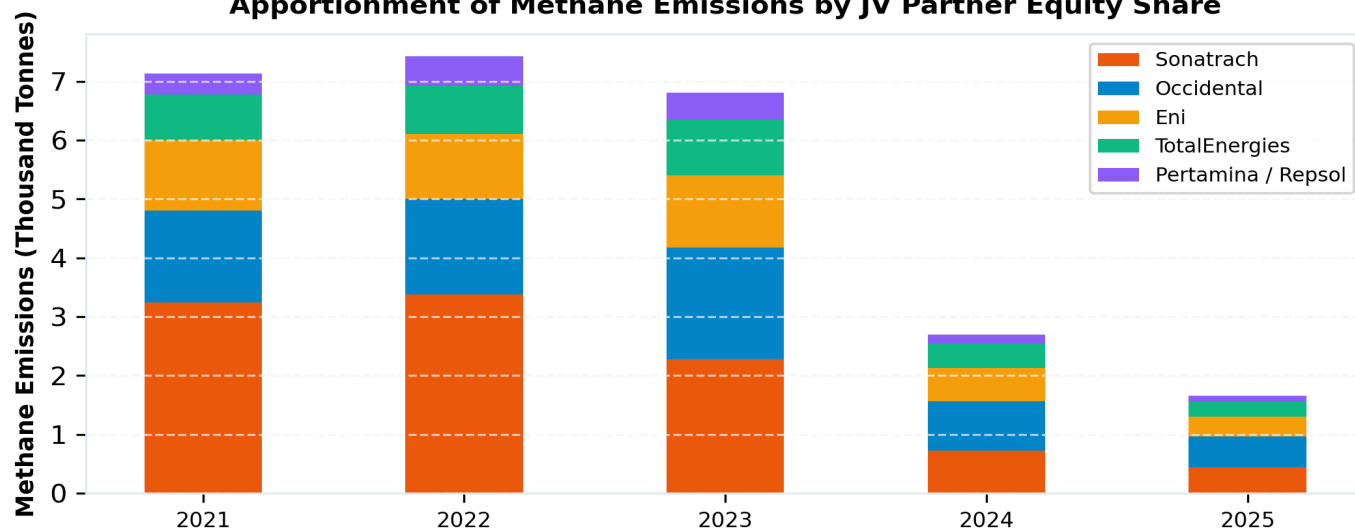
- **Total Production Intensity:** Reached **14.58 kg CO2e / Total BOE**, substantially outperforming the international OGCI Upstream Benchmark ceiling of **17.0 kg CO2e / BOE**.
- **Saleable Production Intensity:** Decreased to **33.72 kg CO2e / Saleable BOE** (-6.2% improvement vs 2021).
- **NGSI Methane Intensity:** Fell to **0.041 wt.%**, positioning Groupement Berkine in the global top decile of upstream producers.

SECTION 9: JV PARTNER EQUITY SHARE ALLOCATIONS

Apportionment of Verified GHG Emissions by JV Partner Equity Share



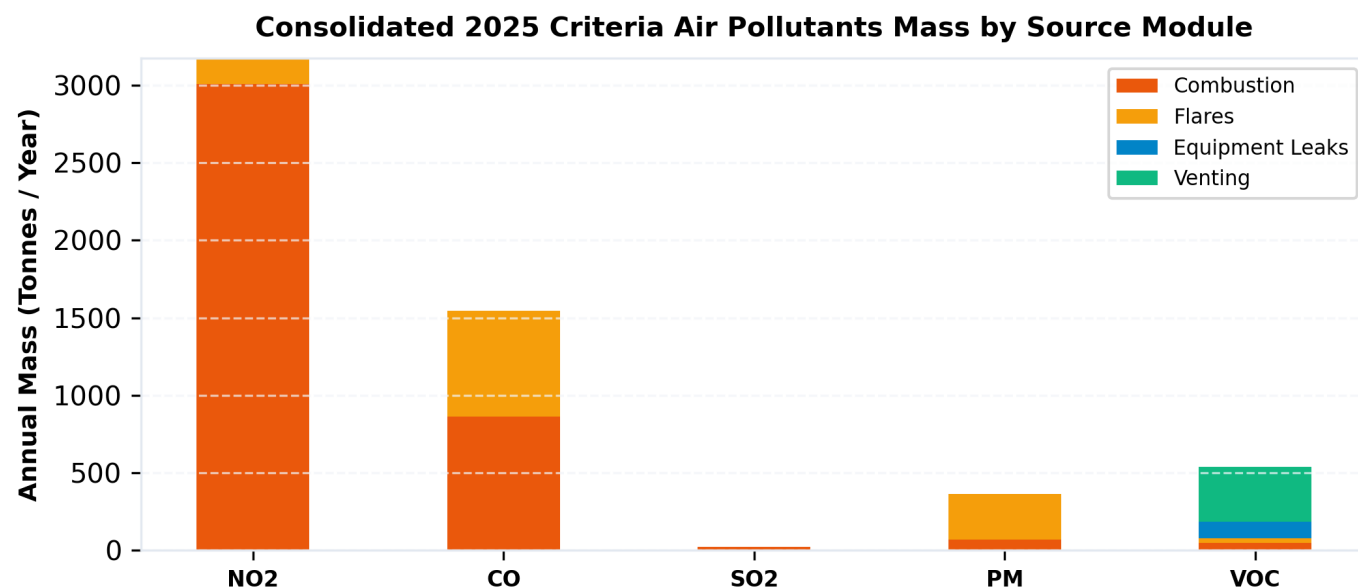
Apportionment of Methane Emissions by JV Partner Equity Share



Corporate Apportionment of GHG and Methane:

In accordance with GHG Protocol Equity Share guidelines, 2025 emissions are allocated across consortium partners based on contractual working interests: Sonatrach: 510,456 tCO₂e (443 tCH₄); Occidental: 595,572 tCO₂e (517 tCH₄); Eni: 382,461 tCO₂e (343 tCH₄); TotalEnergies: 297,786 tCO₂e (258 tCH₄); Pertamina: 79,865 tCO₂e (62 tCH₄); Repsol: 42,744 tCO₂e (33 tCH₄).

SECTION 10: CRITERIA AIR POLLUTANTS (DECREE 06-138)



Criteria Air Pollutant (CAP) Quantification & Concentration Limits:

Under Executive Decree 06-138, atmospheric emissions are regulated both by mass and stack concentration (mg/Nm³). In 2025, total criteria emissions comprised **3,175.5 t NO₂**, **1,544.2 t CO**, **20.4 t SO₂**, **361.2 t PM**, and **536.8 t VOC**. Stack testing verified full compliance for CO (105-118 mg/Nm³ vs 150 limit), SO₂ (12-15 mg/Nm³ vs 800 limit), PM (8.5-11.2 mg/Nm³ vs 30 limit), and VOC (42-52 mg/Nm³ vs 150 limit). Gas turbine NO₂ emissions require dry low-NO_x burner retrofits currently underway.

SECTION 11 & 12: QA/QC VERIFICATION & STATUTORY REGIMES

1. Quality Assurance / Quality Control Verification Protocols

All 2021–2025 activity data, fuel gas chromatography, and custody transfer logs are maintained on tamper-evident digital records. Ultrasonic flare meters undergo semi-annual calibration in accordance with API MPMS Chapter 14. Gas samples are analyzed weekly via on-site gas chromatography to verify high heating values (HHV) and stoichiometric C1-C6+ combustion constants.

2. National & International Statutory Frameworks

Statute / Standard	Regulatory Mandate & Scope
Executive Decree 21-330 Art. 9	Imposes mandatory ceiling on gas flaring not to exceed 1.00% of gross gas production.
Executive Decree 06-138	Regulates atmospheric emission limit values (mg/Nm ³) for industrial installations.
Hydrocarbon Law 19-13	National legal regime governing exploration, development, and environmental protection in Algeria.
ISO 14064-1:2018	International specification for quantification and reporting of greenhouse gas emissions.
API Compendium 2021	Industry standard methodologies and emission factors for oil and gas facilities.
OGMP 2.0 Level 4/5	United Nations Environment Programme framework for empirical methane quantification.

APPENDICES: CONSOLIDATED DATA TABLES (2021–2025)

Complete Historical Production, Modular Emissions, Flaring Streams & Intensities

Table A.1: HBNS Production Profile (2021–2025) [Reference Page 24]

Products	Unit	2021	2022	2023	2024	2025
Gross Gas Production	MMSm ³	5,088.03	5,497.26	5,541.74	5,723.88	5,517.62
Gross Gas Production	MMBOE	42.26	45.62	46.16	47.68	45.96
Gas w/o Injected Gas	MMSm ³	765.51	808.06	772.85	544.06	392.69
Gas w/o Injected Gas	MMBOE	6.36	6.71	6.44	4.53	3.27
Liquid Production (Crude/LPG)	MMBOE	24.85	26.56	24.98	22.17	24.20
Total Production (Gross)	MMBOE	67.11	72.18	71.15	69.85	70.16
Total Production w/o Injected Gas	MMBOE	31.21	33.27	31.42	26.70	27.47
Total Saleable Production	MMBOE	28.36	30.45	28.77	24.50	25.44

Table A.2: El Merk Production Profile (2021–2025) [Reference Page 25]

Products	Unit	2021	2022	2023	2024	2025
Gross Gas Production	MMSm ³	3,655.66	3,964.74	4,244.53	4,353.68	4,239.89
Gross Gas Production	MMBOE	26.27	28.10	29.75	30.46	29.58
Gas w/o Injected Gas	MMSm ³	338.35	380.95	369.26	369.25	374.81
Gas w/o Injected Gas	MMBOE	2.43	2.71	2.59	2.58	2.62
Liquid Production (Crude/LPG)	MMBOE	32.92	32.60	34.99	33.97	31.18
Total Production (Gross)	MMBOE	59.19	60.71	64.74	64.43	60.77
Total Production w/o Injected Gas	MMBOE	35.35	35.32	37.58	36.56	33.80
Total Saleable Production	MMBOE	32.92	32.60	34.99	33.97	31.18

Table A.3: HBNS GHG Emissions by SANGEA Module (tCO₂e) [Reference Page 26]

SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	692,694.21	712,520.69	715,852.19	552,528.86	540,740.16
Flares	284,806.75	244,051.90	176,154.42	202,653.29	138,247.85
Equipment Leaks (Fugitives)	35,072.07	33,940.36	33,890.67	1,056.95	2,494.63
Oil and Gas Venting	12,502.08	13,485.24	13,484.33	15,597.37	15,035.33
Mobile & Transportation	1,512.71	1,949.13	5,839.58	5,838.38	7,293.94
Storage Tanks	1,323.22	1,582.79	1,627.10	1,931.19	2,112.50
Miscellaneous Sources	52.61	19.05	27.40	25.07	19.26
Scope 1 Direct Emissions	1,027,963.65	1,007,549.16	946,875.71	779,631.13	705,943.66
Scope 2 Indirect Emissions	87,220.34	86,435.65	91,303.88	75,803.51	78,086.17
Total GHG Emissions (HBNS)	1,115,183.99	1,093,984.81	1,038,179.59	855,434.64	784,029.83

Table A.4: EI Merk GHG Emissions by SANGEA Module (tCO₂e) [Reference Page 27]

SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	557,093.77	585,340.12	720,643.74	695,899.51	694,054.62
Flares	280,697.29	341,868.29	195,795.53	166,779.72	167,767.37
Equipment Leaks (Fugitives)	68,237.89	70,327.44	71,256.55	1,704.77	5,477.63
Oil and Gas Venting	10,294.91	11,119.57	11,916.70	13,686.82	13,329.14
Mobile & Transportation	1,302.89	2,356.07	3,306.52	2,752.18	3,347.15
Storage Tanks	1,386.77	285.47	1,489.59	1,892.83	1,639.25
Miscellaneous Sources	17.15	10.89	10.10	9.40	13.69
Scope 1 Direct Emissions	919,030.66	1,011,307.84	1,004,418.73	882,725.23	885,628.85
Scope 2 Indirect Emissions	171,947.77	214,554.13	244,599.51	250,473.10	239,226.36
Total GHG Emissions (ELM)	1,090,978.43	1,225,861.97	1,249,018.24	1,133,198.34	1,124,855.22

Table A.5: Groupement Berkine Total GHG by Module (tCO2e) [Reference Page 28]

SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	1,249,787.98	1,297,860.81	1,436,495.93	1,248,428.37	1,234,794.79
Flares	565,504.04	585,920.19	371,949.95	369,433.01	306,015.22
Equipment Leaks (Fugitives)	103,309.96	104,267.80	105,147.22	2,761.72	7,972.26
Oil and Gas Venting	22,796.99	24,604.81	25,401.03	29,284.19	28,364.47
Mobile & Transportation	2,815.60	4,305.20	9,146.10	8,590.55	10,641.09
Storage Tanks	2,709.99	1,868.26	3,116.69	3,824.02	3,751.75
Miscellaneous Sources	69.76	29.94	37.50	34.47	32.95
Scope 1 Direct Emissions	1,946,994.31	2,018,857.00	1,951,294.44	1,662,356.36	1,591,572.51
Scope 2 Indirect Emissions	259,168.11	300,989.78	335,903.39	326,276.62	317,312.54
Total of GHG Emissions	2,206,162.42	2,319,846.78	2,287,197.83	1,988,632.97	1,908,885.05

Table A.6: HBNS Methane (CH4) Emissions by Module (Tonnes) [Reference Page 29]

SANGEA Module	2021	2022	2023	2024	2025
Stationary Combustion	38.12	36.88	37.16	33.40	34.30
Flares	899.16	770.20	573.21	653.33	37.08
Equipment Leaks	1,401.76	1,356.56	1,354.57	37.75	89.09
Oil and Gas Venting	499.62	538.92	538.89	556.60	536.54
Mobile & Transportation	0.39	1.27	1.53	1.01	1.15
Storage Tanks	52.93	63.31	65.08	68.97	75.45
Scope 1 Methane	2,891.98	2,767.15	2,570.45	1,351.06	773.62
Scope 2 Methane	9.98	9.89	10.45	8.72	8.99
Total Methane (HBNS)	2,901.97	2,777.04	2,580.90	1,359.78	782.61

Table A.7: El Merk Methane (CH₄) Emissions by Module (Tonnes) [Reference Page 30]

SANGIA Module	2021	2022	2023	2024	2025
Stationary Combustion	31.18	34.49	42.20	40.61	39.82
Flares	988.72	1,328.14	781.30	656.52	76.49
Equipment Leaks	2,724.10	2,807.79	2,844.54	60.88	195.63
Oil and Gas Venting	403.32	443.89	475.65	487.88	475.13
Mobile & Transportation	0.38	1.28	1.44	0.86	0.94
Storage Tanks	55.47	11.42	59.58	67.60	58.54
Scope 1 Methane	4,203.17	4,627.02	4,204.71	1,314.36	846.56
Scope 2 Methane	19.68	24.56	28.00	28.82	27.53
Total Methane (ELM)	4,222.85	4,651.57	4,232.70	1,343.18	874.08

Table A.8: Groupement Berkine Total Methane by Module (Tonnes) [Reference Page 31]

SANGIA Module	2021	2022	2023	2024	2025
Stationary Combustion	69.30	71.37	79.36	74.01	74.12
Flares	1,887.88	2,098.34	1,354.51	1,309.85	113.57
Equipment Leaks	4,125.86	4,164.35	4,199.11	98.63	284.72
Oil and Gas Venting	902.94	982.81	1,014.54	1,044.48	1,011.68
Mobile & Transportation	0.77	2.55	2.97	1.87	2.10
Storage Tanks	108.40	74.73	124.66	136.57	133.99
Scope 1 Methane	7,095.15	7,394.17	6,775.16	2,665.41	1,620.18
Scope 2 Methane	29.66	34.45	38.45	37.54	36.51
Total Methane (GB)	7,124.82	7,428.61	6,813.60	2,702.96	1,656.69

Table A.9: Consolidated CPF Flaring Streams (kNm³) [Reference Page 32]

Asset & Year	Routine Flaring	Non-Routine Flaring	Safety Flaring	Total CPF Flared
HBNS 2021	25,428.49	62,429.83	1,819.94	89,678.26
HBNS 2022	26,765.98	34,736.81	1,932.68	63,435.47
HBNS 2023	26,842.94	21,860.97	1,959.52	50,663.43
HBNS 2024	26,450.85	35,233.83	1,905.84	63,590.52
HBNS 2025	26,819.69	18,971.93	1,959.52	47,751.14
ELM 2021	29,081.24	37,938.84	29,365.70	96,385.78
ELM 2022	43,816.61	72,950.32	5,256.74	122,023.67
ELM 2023	46,806.84	18,628.80	4,225.82	69,661.46
ELM 2024	46,814.62	8,779.38	4,237.40	59,831.40
ELM 2025	44,786.50	16,369.70	4,225.82	65,382.02

Table A.10: HBNS 5-Year Performance Ledger (2021–2025) [Reference Page 33]

Key Metric	Unit	2021	2022	2023	2024	2025
Scope 1 GHG Direct	MMtCO2e	1.028	1.008	0.947	0.780	0.706
Scope 2 GHG Indirect	MMtCO2e	0.087	0.086	0.091	0.076	0.078
Total GHG Emissions	MMtCO2e	1.115	1.094	1.038	0.855	0.784
Methane Mass (CH4)	tCH4	2,901.97	2,777.04	2,580.90	1,359.78	782.61
Total Flare Volume	MMSm³	96.965	83.192	61.099	69.638	51.339
Total Production	MMBOE	67.111	72.183	71.148	69.853	70.160
Scope 1 GHG Intensity	tCO2e/BOE	0.0153	0.0140	0.0133	0.0112	0.0101
Total GHG Intensity	tCO2e/BOE	0.0166	0.0152	0.0146	0.0122	0.0112
Flaring Intensity (% Gas)	vol. %	1.91%	1.51%	1.10%	1.22%	0.93% (PASS)
Total Saleable Production	MMBOE	28.360	30.453	28.773	24.497	25.435
Saleable GHG Intensity	tCO2e/BOE	0.0393	0.0359	0.0361	0.0349	0.0308
NGSI Methane Intensity	wt. %	0.166%	0.149%	0.147%	0.092%	0.051%

Table A.11: El Merk 5-Year Performance Ledger (2021–2025) [Reference Page 34]

Key Metric	Unit	2021	2022	2023	2024	2025
Scope 1 GHG Direct	MMtCO ₂ e	0.919	1.011	1.004	0.883	0.886
Scope 2 GHG Indirect	MMtCO ₂ e	0.172	0.215	0.245	0.250	0.239
Total GHG Emissions	MMtCO ₂ e	1.091	1.226	1.249	1.133	1.125
Methane Mass (CH ₄)	tCH ₄	4,222.85	4,651.57	4,232.70	1,343.18	874.08
Total Flare Volume	MMSm ³	101.682	129.575	75.558	63.344	70.148
Total Production	MMBOE	59.192	60.709	64.742	64.429	60.767
Scope 1 GHG Intensity	tCO ₂ e/BOE	0.0155	0.0167	0.0155	0.0137	0.0146
Total GHG Intensity	tCO ₂ e/BOE	0.0184	0.0202	0.0193	0.0176	0.0185
Flaring Intensity (% Gas)	vol. %	2.78%	3.27%	1.78%	1.45%	1.65%
Total Saleable Production	MMBOE	32.922	32.605	34.993	33.971	31.184
Saleable GHG Intensity	tCO ₂ e/BOE	0.0331	0.0376	0.0357	0.0334	0.0361
NGSI Methane Intensity	wt. %	0.164%	0.176%	0.148%	0.047%	0.033%

Table A.12: Groupement Berkine Consolidated Performance Ledger (2021–2025) [Reference Page 35]

Key Metric	Unit	2021	2022	2023	2024	2025
Scope 1 GHG Direct	MMtCO ₂ e	1.947	2.019	1.951	1.662	1.592
Scope 2 GHG Indirect	MMtCO ₂ e	0.259	0.301	0.336	0.326	0.317
Total GHG Emissions	MMtCO ₂ e	2.206	2.320	2.287	1.989	1.909
Methane Mass (CH ₄)	tCH ₄	7,124.82	7,428.62	6,813.60	2,702.96	1,656.69
Total Flare Volume	MMSm ³	196.296	212.767	136.657	132.982	121.488
Routine Flaring	MMSm ³	100.397	83.367	81.771	77.291	75.538
Non-Routine Flaring	MMSm ³	3.370	2.682	3.425	46.432	37.282
Safety & Purge Flaring	MMSm ³	94.880	126.718	51.461	6.481	6.525
Total Production (Gross)	MMBOE	126.303	132.893	135.890	134.282	130.927
Total GHG Intensity	tCO ₂ e/BOE	0.0175	0.0175	0.0168	0.0148	0.0146
Flaring Intensity (% Gas)	vol. %	2.25%	2.25%	1.40%	1.32%	1.25%
Total Saleable Production	MMBOE	61.282	63.058	63.766	58.468	56.619
Saleable GHG Intensity	tCO ₂ e/BOE	0.0360	0.0368	0.0359	0.0340	0.0337
NGSI Methane Intensity	wt. %	0.165%	0.158%	0.147%	0.072%	0.041%

Table A.13: Total GHG (CO₂e) by JV Partner Equity Share (tCO₂e) [Reference Page 36]

JV Partner	HBNS 2021	ELM 2021	Total 2021	HBNS 2024	ELM 2024	Total 2024	HBNS 2025	ELM 2025	Total 2025
Sonatrach	486,220	510,578	996,798	229,256	302,564	531,820	210,120	300,336	510,456
Occidental	233,073	245,470	478,544	266,896	353,558	620,453	244,617	350,955	595,572
Eni	278,796	122,190	400,986	225,835	176,779	402,614	206,984	175,477	382,461
TotalEnergies	117,094	122,190	239,284	133,448	176,779	310,227	122,309	175,477	297,786
Pertamina	0	58,913	58,913	0	80,457	80,457	0	79,865	79,865
Repsol	0	31,638	31,638	0	43,062	43,062	0	42,744	42,744
Total CO ₂ e	1,115,184	1,090,978	2,206,162	855,435	1,133,198	1,988,633	784,030	1,124,855	1,908,885

Table A.14: Total Methane (CH₄) by JV Partner Equity Share (Tonnes) [Reference Page 37]

JV Partner	HBNS 2021	ELM 2021	Total 2021	HBNS 2024	ELM 2024	Total 2024	HBNS 2025	ELM 2025	Total 2025
Sonatrach	1,265	1,976	3,242	364	359	723	210	233	443
Occidental	607	950	1,557	424	419	843	244	273	517
Eni	725	473	1,198	359	210	569	207	136	343
TotalEnergies	305	473	778	212	210	422	122	136	258
Pertamina	0	228	228	0	95	95	0	62	62
Repsol	0	122	122	0	51	51	0	33	33
Total CH ₄	2,902	4,223	7,125	1,360	1,343	2,703	783	874	1,657

Table A.15: HBNS Criteria Air Pollutants by Module (Tonnes/Yr) [Reference Pages 38–39]

Year	Source Module	NO2	CO	SO2	PM	VOC
2021	Stationary Combustion	1,574.03	468.64	8.29	36.86	24.47
2021	Flare Systems	141.17	581.29	0.88	249.12	528.06
2021	Equipment Leaks	—	—	—	—	364.02
2021	Oil & Gas Venting	—	—	—	—	293.41
2021	Total HBNS CAP	1,715.19	1,049.93	9.17	285.98	1,669.18
2023	Stationary Combustion	1,550.16	484.61	8.67	39.12	26.51
2023	Flare Systems	87.44	360.03	0.56	154.30	303.44
2023	Equipment Leaks	—	—	—	—	717.06
2023	Oil & Gas Venting	—	—	—	—	285.27
2023	Total HBNS CAP	1,637.60	844.64	9.23	193.42	1,332.28
2025	Stationary Combustion	1,340.53	363.13	5.87	26.01	15.85
2025	Flare Systems	74.18	305.44	0.47	130.90	19.63
2025	Equipment Leaks	—	—	—	—	86.87
2025	Oil & Gas Venting	—	—	—	—	284.03
2025	Total HBNS CAP	1,414.70	668.58	6.35	156.91	406.37

Table A.16: EI Merk Criteria Air Pollutants by Module (Tonnes/Yr) [Reference Pages 40–41]

Year	Source Module	NO2	CO	SO2	PM	VOC
2021	Stationary Combustion	1,306.03	396.94	10.18	34.29	24.63
2021	Flare Systems	138.70	571.12	0.93	244.76	364.07
2021	Equipment Leaks	—	—	—	—	373.48
2021	Oil & Gas Venting	—	—	—	—	53.46
2021	Total ELM CAP	1,444.73	968.06	11.11	279.06	815.64
2023	Stationary Combustion	1,727.01	510.26	11.62	41.71	28.69
2023	Flare Systems	96.74	398.33	0.69	170.71	134.90
2023	Equipment Leaks	—	—	—	—	491.14
2023	Oil & Gas Venting	—	—	—	—	82.13
2023	Total ELM CAP	1,823.75	908.59	12.31	212.42	736.86
2025	Stationary Combustion	1,669.62	500.29	13.43	43.39	31.28
2025	Flare Systems	91.16	375.36	0.65	160.87	11.11
2025	Equipment Leaks	—	—	—	—	19.03
2025	Oil & Gas Venting	—	—	—	—	69.03
2025	Total ELM CAP	1,760.78	875.66	14.08	204.26	130.45

EXECUTIVE VERIFICATION SIGN-OFF & CERTIFICATION

This quantified annual greenhouse gas and criteria air pollutants inventory has been prepared in accordance with the mandatory specifications of ISO 14064-1:2018, the GHG Protocol Corporate Standard, and Algerian Executive Decrees 21-330 and 06-138. The reported activity data, gas chromatography compositions, continuous optical gas imaging DRE measurements, and emissions estimates represent a true, fair, and reconciled view of Groupement Berkine environmental performance for the fiscal year 2025.

All historical records from 2021 through 2025 have been verified against original Central Processing Facility (CPF) custody transfer meters, DCS flow computers, and approved joint venture partner equity allocation contracts.

Lead MRV & Carbon Accounting Engineer:	Managing Director:
HSE & Decarbonization Division Groupement Berkine	Executive Management Committee Sonatrach & International Partners